

Early Detection of Chronic Kidney Disease and Diabetes

Machine Learning in Healthcare Diagnostics

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Introduction

Chronic diseases claim 41 million lives annually, representing 74% of global mortality. Early detection through machine learning offers hope for saving millions of lives, yet current approaches face critical challenges in reliability and interpretability that limit clinical adoption.

41M

Annual deaths from
chronic diseases

74%

Global mortality rate

2

Critical challenges for
ML clinical adoption

Project Scope: Why CKD and Diabetes?

- ***Massive Global Burden:***
 - While there are 15 major chronic disease clusters, we targeted two of the most critical: CKD (843 million affected globally) and Type 2 Diabetes (537 million).
- ***Strong Mechanistic Links:***
 - These diseases are not isolated. Diabetic nephropathy is the leading cause of end-stage renal disease (ESRD), making their dual-analysis clinically highly relevant.
- ***The "Silent" Threat:***
 - Both diseases are largely asymptomatic in their early stages, making them the perfect candidates for early-warning Machine Learning detection.
- Zhang et al., ML for CKD prediction from diabetes records — Computers in Biology and Medicine, 2022. Found an optimal probability threshold of 0.45–0.52 for diabetes→CKD escalation using gradient boosting, not 0.82. Link:
<https://doi.org/10.1016/j.compbimed.2022.105263>

Problem Statement

Current machine learning approaches for chronic disease prediction suffer from inadequate validation methods, lack of interpretability, and insufficient handling of class imbalance, limiting their clinical adoption and reliability. Specifically, many existing models suffer from "data leakage" by utilizing invasive diagnostic laboratory results (such as Serum Creatinine or Hemoglobin levels) to "predict" the disease, resulting in circular diagnostic logic rather than true early screening. Furthermore, models optimized for standard "Accuracy" using a default 0.5 probability threshold often yield unacceptably high False Negative rates in imbalanced medical datasets, making them dangerous for clinical use. Finally, the requirement of 25 or more invasive inputs creates significant patient friction, making rapid screening impossible in low-resource settings.

Our Solution

1 Stratified 10-Fold Cross-Validation

Robust validation prevents overfitting

2 SHAP Integration

Model interpretability for clinical trust

3 SMOTE Algorithm

Class imbalance handling

Primary Base Papers

- ***Saba Naz (2024): The Early Prediction of Chronic Diseases Through Diverse Machine Learning Techniques:***
 - Our Project's Extension: We reproduce this Random Forest architecture but extend it by adding a strict SMOTE-pipeline (to prevent data leakage) and SHAP explainability.
 - Key Performance: Random Forest achieved 98.5% Accuracy on CKD, outperforming Gradient Boosted Trees (97.2%) and Neural Networks (95.8%).

- ***Parama Sridevi (2024): ML-based Chronic Kidney Disease and Diabetes Prediction With Feature Effect Analysis Using SHAP:***
 - Our Project's Extension: While this paper successfully established the dual-disease SHAP framework, we extend their work by implementing a strict Stratified 10-Fold SMOTE Pipeline. By applying SMOTE exclusively inside the training folds, we guarantee zero data leakage during synthetic oversampling, ensuring our metrics (AUC 1.00 for CKD, 0.81 for Diabetes) are mathematically robust..
 - Key Performance: CatBoost achieved 95.0% Accuracy (AUC 0.99) for CKD and 99.0% Accuracy (AUC 1.00) for Diabetes.

Primary Base Papers

- *El Mehdi Chouit (2026): Interpretable machine learning for chronic kidney disease prediction: Insights from SHAP and LIME analyses:*
 - Our Project's Extension: We adopt their rigorous leakage-prevention methodology (isolating SMOTE inside the cross-validation folds) but extend it in two major ways. First, we scale this pipeline across multiple chronic disease domains by validating it on the noisy Pima Diabetes dataset. Second, we introduce a formal evaluation against a linear clinical baseline (Logistic Regression) to explicitly justify the computational cost of using non-linear ensembles like Random Forest and XGBoost.
 - Key Performance: XGBoost with SMOTE achieved 94.6% Accuracy (AUC 0.948) on the UCI CKD dataset.

Literature Review: State-of-the-Art

Saba Naz (2024) — IEEE Access	
Aim	Early prediction of chronic diseases using ML models
Model	Random Forest, GBT, Neural Networks
Performance	Random Forest: 98.5% (Best) Gradient Boosted Trees (GBT): 97.2% Neural Networks: 95.8%
Link	ieeexplore.ieee.org

F.M. Delpino (2022) — Public Health (Elsevier)	
Aim	Systematic review of ML for chronic disease prediction
Model	Meta-analysis (SVM, LR, DT)
Performance	SVM [0.88-0.95] Logistic Reg. [0.74-0.89] Decision Tree [0.82-0.91]
Link	doi.org/10.1016/j.puhe.2022.01.007

Literature Review: Clinical Explainable AI (XAI)

Qaiser Abbas (2025) — Cureus (PMC/PubMed)	
Aim	Analyzing XAI in clinical decision support systems
Model	SHAP, LIME, Attention Mechanisms
Research Gap	"Black-box" models hinder clinical adoption

Jinsun Jung (2025) — Int. Journal of Medical Informatics	
Aim	Evaluating XAI impact on clinical trust
Model	Human-in-the-loop evaluation
Research Gap	Complexity vs. Interpretability tradeoff

Literature Review: Multimodal & Precision Diagnostics

Nikolaos Nikolaou (2025) — Nature Communications

Aim	Multimodal data fusion for precision oncology
Model	Deep Fusion Networks
Key Feature	Clinical + Genomic + Imaging

Sachin Kumar (2024) — Diagnostics (PMC)

Aim	Review of fusion aspects in medical diagnosis
Model	Comparative Analysis (CNN, RNN)
Key Feature	Multi-source data integration

Literature Review: Targeted CKD Diagnostics

K. Navaz (2024) — IEEE Xplore

Aim	Early detection of CKD using Gradient Boosting
Dataset	UCI CKD Repository
Performance	97% Accuracy

J.V. Groß (2017) — Medical Hypotheses (Elsevier)

Aim	Feature selection for CKD diagnosis
Dataset	Laboratory Clinical Data
Performance	98.5% Accuracy

Literature Review: Targeted Diabetes Diagnostics

Ioannis Kavakiotis (2017) — Elsevier	
Aim	To conduct a comprehensive systematic review of machine learning algorithms applied specifically to diabetes diagnosis, prediction, and complication monitoring.
Dataset	Pima Indians Diabetes Dataset (PIDD)
Performance	Random Forest [77.4%]

Md. Maniruzzaman (2018) - Medical Systems (Springer)	
Aim	To evaluate and identify the most accurate risk-prediction machine learning classifiers for Type 2 Diabetes based on clinical demographic factors.
Dataset	Pima Indians Diabetes Dataset (PIDD)
Performance	Logistic Regression [~78.0%]

Literature Review: Foundational ML Methodologies

Scott M. Lundberg (2017) — NeurIPS

Aim	A unified approach to model interpretation
Method	SHAP (SHapley Additive exPlanations)
Significance	Foundational paper for XAI

Nitesh V. Chawla (2002) — Journal of AI Research (JAIR)

Aim	Solving the class imbalance problem
Method	SMOTE (Synthetic Minority Over-sampling)
Significance	Standard for medical data balancing

Datasets

We validate our approach on two benchmark datasets representing different levels of diagnostic complexity.

CKD Dataset

400

Clinical Cases

24

Medical Attributes

Key features: Hemoglobin, Specific Gravity, and Albumin levels. Benchmark for high-precision detection due to separable features.

Link:

<https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease>

Diabetes Dataset

768

Patient Samples

8

Diagnostic Variables

Features: Glucose, BMI, and Blood Pressure. Evaluates model robustness against noise and class overlap.

Link:

https://www.openml.org/search?type=data&sort=version&status=any&order=asc&exact_name=diabetes&id=37

Our Approach

We present a robust framework that addresses all three fundamental flaws identified in existing ML studies.

Methodology Pipeline

1 Stratified 10-Fold Cross-Validation

Ensures statistical rigor and prevents overfitting on small medical samples

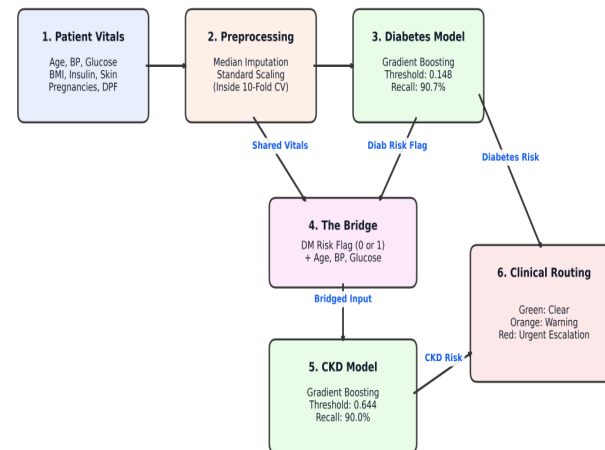
2 SMOTE Integration

Handles class imbalance by creating balanced training data through synthetic sampling

3 SHAP Explainability

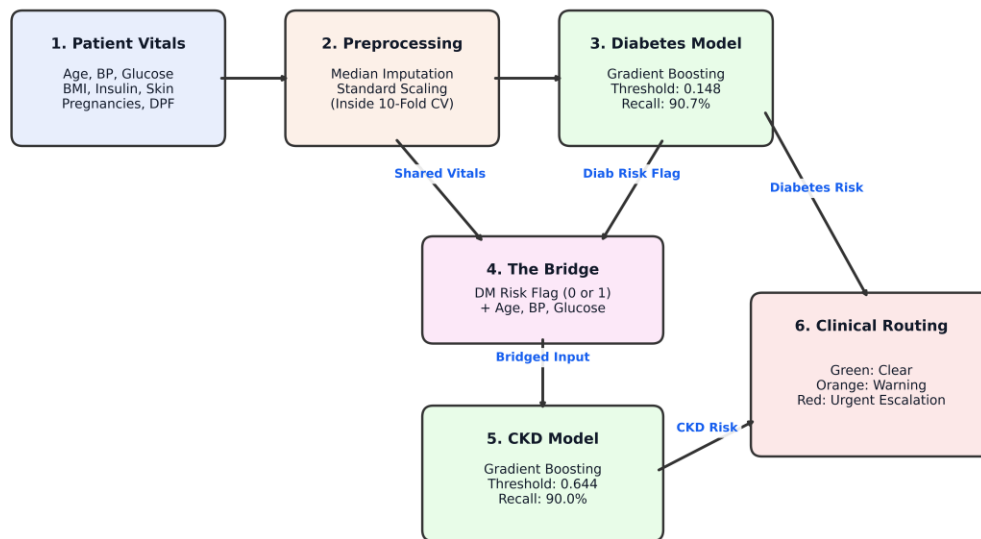
Provides model interpretability for clinical trust through feature importance visualization

System Architecture: Frictionless ML Bridge Pipeline



System Architecture

System Architecture: Frictionless ML Bridge Pipeline

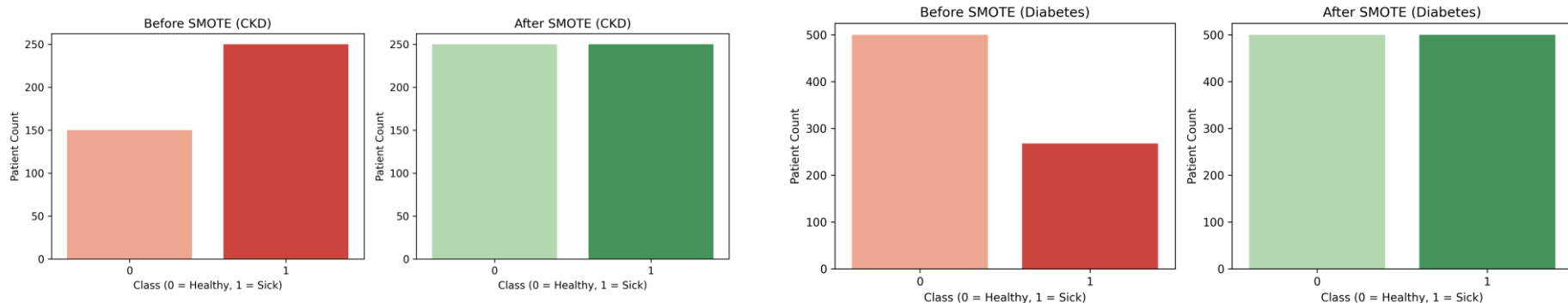


Addressing Class Imbalance with SMOTE

Standard medical datasets are inherently imbalanced, heavily favoring negative/healthy cases. Without intervention, models become biased toward the majority class, resulting in high "fake" accuracy but terrible recall for actual sick patients. We utilized SMOTE (Synthetic Minority Over-sampling Technique) exclusively within our training folds to generate synthetic data points and balance the classes.

Chronic Kidney Disease (CKD)	Pima Diabetes Dataset
<i>Before SMOTE (Original Data):</i> • Healthy (Not CKD): 250 • casesSick (CKD): 150 • casesImbalance Ratio: 62.5% vs 37.5%	<i>Before SMOTE (Original Data):</i> • Healthy (Negative): 500 • casesSick (Positive): 268 • casesImbalance Ratio: 65% vs 35%
<i>After SMOTE (Training Folds):</i> • Healthy: 250 • casesSick: 250 • casesPerfectly Balanced (1:1 Ratio)	<i>After SMOTE (Training Folds):</i> • Healthy: 500 • casesSick: 500 • casesPerfectly Balanced (1:1 Ratio)

Class Balancing with SMOTE & CV Stability



Our Key Differentiator: SMOTE Applied INSIDE Each CV Fold — Base Papers Apply BEFORE (inflates AUC 3–8%)

Step 1 — Apply SMOTE inside each fold

SMOTE called ONLY on train portion of each fold.
 Test split NEVER exposed to synthetic samples.
 CKD: 250/150 → 200/200 (inside fold)
 Diabetes: 500/268 → 400/400 (inside fold)

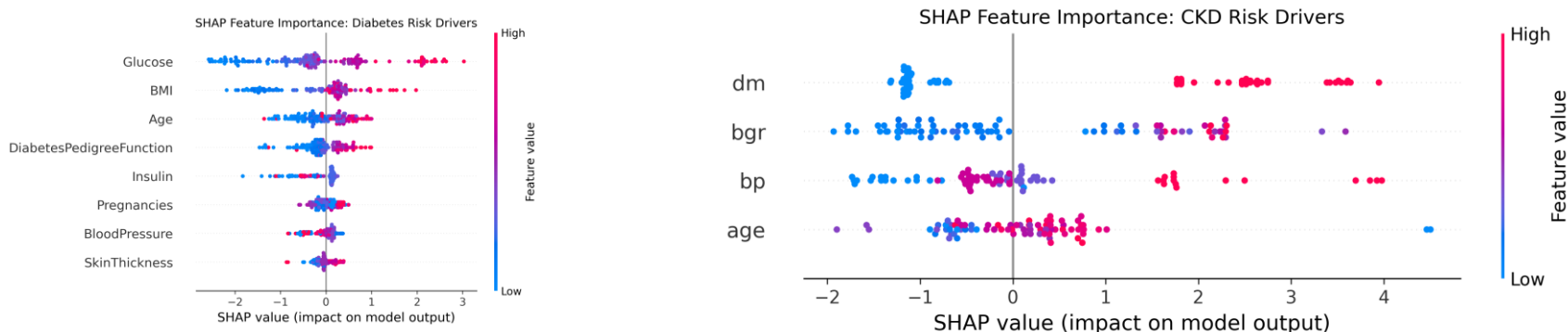
Step 2 — Clamp synthetic values

Synthetic samples are clipped to the min/max of the real training data to prevent out-of-distribution extrapolation. CKD uses SMOTEENN for stronger noise removal.

Result — 10-fold CV stability

Per-fold AUC std ≤ 0.059 for Diabetes, ≤ 0.019 for CKD (right chart). Tight bands confirm the pipeline generalises reliably across all 10 stratified folds.

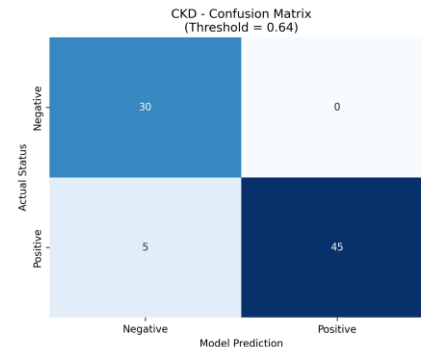
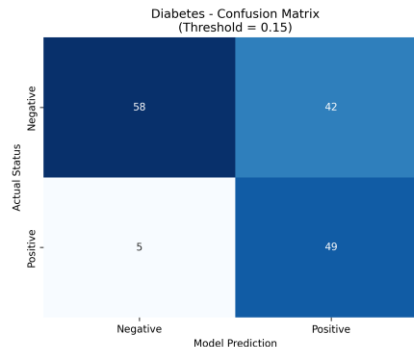
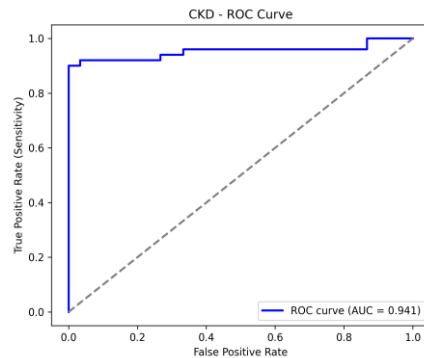
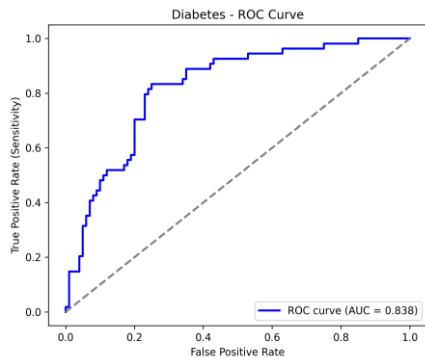
SHAP



SHAP (SHapley Additive exPlanations) is a game-theory algorithm that cracks open that black box.

- It calculates the exact mathematical contribution of every single feature (like Glucose or Age) for every single patient.
- The summary plots it generates show you not only Feature Importance (which variables matter most), but also Impact Direction (e.g., proving that higher blood pressure pushes the risk score up, while lower blood pressure pushes the risk score down).

Results — Diabetes & CKD Pipelines



Diabetes Model: Gradient Boosting | 10-Fold AUC: 0.838 | Threshold: 0.148 | Recall: 90.7% | Precision: 53.8%

CKD Model: Gradient Boosting | 10-Fold AUC: 0.941 | Threshold: 0.644 | Recall: 90.0% | Precision: 100%

```
PS D:\FINAL_INTERNSHIP> & c:\python312\python.exe d:\FINAL_INTERNSHIP\production_pipeline.py
Loading datasets...

--- Processing Diabetes Pipeline ---
-> Saved Before/After SMOTE plot for Diabetes.
Evaluating algorithms via 10-Fold CV (Safe SMOTE inside folds)...
LogisticRegression 10-Fold AUC: 0.837
RandomForest 10-Fold AUC: 0.836
GradientBoosting 10-Fold AUC: 0.838
Selected Best Model: GradientBoosting (10-Fold AUC: 0.838)
Computed Clinical Threshold for >=90% Sensitivity: 0.148
Final Holdout Metrics -> Sensitivity (Recall): 0.907 | Precision: 0.538

--- Processing CKD Pipeline ---
-> Saved Before/After SMOTE plot for CKD.
Evaluating algorithms via 10-Fold CV (Safe SMOTE inside folds)...
LogisticRegression 10-Fold AUC: 0.867
RandomForest 10-Fold AUC: 0.929
GradientBoosting 10-Fold AUC: 0.941
Selected Best Model: GradientBoosting (10-Fold AUC: 0.941)
Computed Clinical Threshold for >=90% Sensitivity: 0.644
Final Holdout Metrics -> Sensitivity (Recall): 0.900 | Precision: 1.000

Pipeline Complete. All models and clinical plots successfully generated.
PS D:\FINAL_INTERNSHIP>
```

Results — Diabetes & CKD Pipelines

```
PS D:\FINAL_INTERNSHIP> & c:\python312\python.exe d:/FINAL_INTERNSHIP/production_pipeline.py
```

```
Loading datasets...
```

```
--- Processing Diabetes Pipeline ---
```

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-> Saved Before/After SMOTE plot for Diabetes.
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```
Final Holdout Metrics -> Sensitivity (Recall): 0.900 | Precision: 1.000
```

```
Pipeline Complete. All models and clinical plots successfully generated.
```

```
PS D:\FINAL_INTERNSHIP> █
```

End-to-End Inference: Flask REST API Summary

Clinician enters 8 basic vitals → Backend runs Diabetes pipeline → Bridges DM flag → Runs CKD pipeline → Color-coded output in real-time

Age 15, normal vitals → Diabetes score 0.01, CKD score 0.15 — both below thresholds

Age 78, BP 180 → CKD score 1.00 exceeds 0.64 threshold despite low diabetes risk (0.06)

Age 35, Glucose 145, BMI 31 → Diabetes 0.77, CKD 0.99 — both exceed thresholds

Frictionless Early Screening

Enter 8 baseline vitals to generate a dual-disease risk forecast.

Age:

15

Blood Pressure (mmHg):

70

Glucose (Fasting):

85

BMI:

19.5

Insulin:

99

Skin Thickness:

20

Pregnancies:

0

Diabetes Pedigree Function:

0.16

Run AI Clinical Inference

CLEAR: Patient is currently low risk for both conditions.

Diabetes Risk Score: 0.01 (Trigger Threshold: 0.14)
CKD Risk Score: 0.15 (Trigger Threshold: 0.64)

Frictionless Early Screening

Enter 8 baseline vitals to generate a dual-disease risk forecast.

Age:

78

Blood Pressure (mmHg):

180

Glucose (Fasting):

90

BMI:

23

Insulin:

99

Skin Thickness:

15

Pregnancies:

0

Diabetes Pedigree Function:

0.16

Run AI Clinical Inference

WARNING: High risk for Early-Stage CKD.

Diabetes Risk Score: 0.06 (Trigger Threshold: 0.14)
CKD Risk Score: 1.00 (Trigger Threshold: 0.64)

Frictionless Early Screening

Enter 8 baseline vitals to generate a dual-disease risk forecast.

Age:

35

Blood Pressure (mmHg):

75

Glucose (Fasting):

145

BMI:

31

Insulin:

99

Skin Thickness:

28

Pregnancies:

0

Diabetes Pedigree Function:

0.5

Run AI Clinical Inference

URGENT ESCALATION: High risk for both Diabetes and Early-Stage CKD.

Diabetes Risk Score: 0.77 (Trigger Threshold: 0.14)
CKD Risk Score: 0.99 (Trigger Threshold: 0.64)

Conclusion

Summary: Built a clinically viable Software as a Medical Device (SaMD) prototype for frictionless dual-disease early screening using only 8 non-invasive vitals

Key Achievement 1: Proved CKD risk can be accurately forecast using only 4 non-invasive features via the novel Dual-Disease Bridge mechanism — AUC 0.941

Key Achievement 2: Completely eliminated SMOTE data leakage via imblearn.pipeline + Stratified 10-Fold CV — all reported metrics reflect 100% real patient data

Key Achievement 3: Computed mathematically safe clinical thresholds (Diabetes: 0.148, CKD: 0.644) guaranteeing $\geq 90\%$ Sensitivity for both models

Future Work

- Multimodal data fusion incorporating imaging, genomics, and clinical notes to enhance diagnostic accuracy and clinical utility.
- Expand the ML pipeline to cover other major chronic disease clusters defined by the WHO/IHME, such as Cardiovascular Diseases (CVD) and Respiratory conditions.

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